

MERADGEN FORTRAN vs. Cpp

100,000 radiative events on a shared random stream

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1 Introduction

MERADGEN is a Møller-scattering radiative-correction event generator. This note compares the live FORTRAN 77 reference to the parity Cpp port that is intended to reproduce it, including FORTRAN’s float32 `sngl` four-vectors. “Parity port” here means we compare FORTRAN and Cpp event-by-event for the parity analysis: same constants, same 32-bit outputs. It is a port check, not a physics upgrade. `meradgen-cpp-final/` is the production tree (PDG constants, double precision throughout, no `sngl`) and is not used in this comparison.

Directory	In this report	Role
<code>meradgen-fortran/</code>	compared	Upstream FORTRAN 77 reference (byte-identical to <code>meradgen10.tar</code>).
<code>meradgen-cpp/</code>	compared	Parity Cpp port: built to match FORTRAN event-by-event for the parity analysis (same constants, 32-bit <code>sngl</code> outputs).
<code>meradgen-cpp-final/</code>	not used	Production library for MolPol/Geant4: PDG constants, double precision, no <code>sngl</code> . Expected to differ from FORTRAN.

1.1 How the 100,000-event sample was built

Each `meradgen` call draws four uniform random numbers (a “quad”)¹ and returns either a non-radiative event (`ich = 0`, no real photon) or a radiative event (`ich = 1`, a real photon). That choice is part of the generator, not something we set by hand. To collect **100,000 radiative events** we therefore had to run many more calls and keep only those with `ich = 1`.

Both directories read the **same** random stream (Python `random.random()`, seed 20260831, 500,000 quads available). FORTRAN and Cpp each processed the stream in order, including non-radiative calls, because `meradgen` keeps internal state from call to call. They both reached the 100,000th radiative event on `meradgen` call **427,348**. That is a radiative fraction of 23.4%. The plots below contain only those 100,000 radiative events.

Kinematics match `meradgen-fortran/run.f`: `elab = 45 GeV`, `thetacm = 90°`, `phi = 10°`, `p1 = -1`. Official compared outputs are the FORTRAN-style float32 four-vectors `VPRAD` and `PHIRAD`. Deltas are Cpp minus FORTRAN, in GeV. Internal KIN doubles (`vgen`, `t1gen`, `zgen`) are not the pass/fail: they can differ without moving the float32 four-vectors. The comparison is event-by-event on those dumped components, not a comparison of histogram bins.

¹MERADGEN consumes the four numbers in order: the first chooses the radiative channel (and, if radiative, samples v); the next two sample t_1 and z ; the fourth sets a photon azimuthal sign. Example: call 1 reads the first line of `quads.txt`, four values in $[0, 1)$ such as 0.1994, 0.1885, 0.5132, 0.4547. The file is written by `comparison/generate_stream.py` (here seed 20260831, 500,000 quads).

2 Overlay spectra

Figure 1 is the physics picture: radiated photon energy E_γ and outgoing CM electron momentum $|k_2|$, reconstructed from VPRAD and PHIRAD. FORTRAN (solid) and Cpp (dashed) fill every bin with the same counts. The radiative channel `ich` never differs. No event exceeds relative 10^{-6} on any four-vector component.

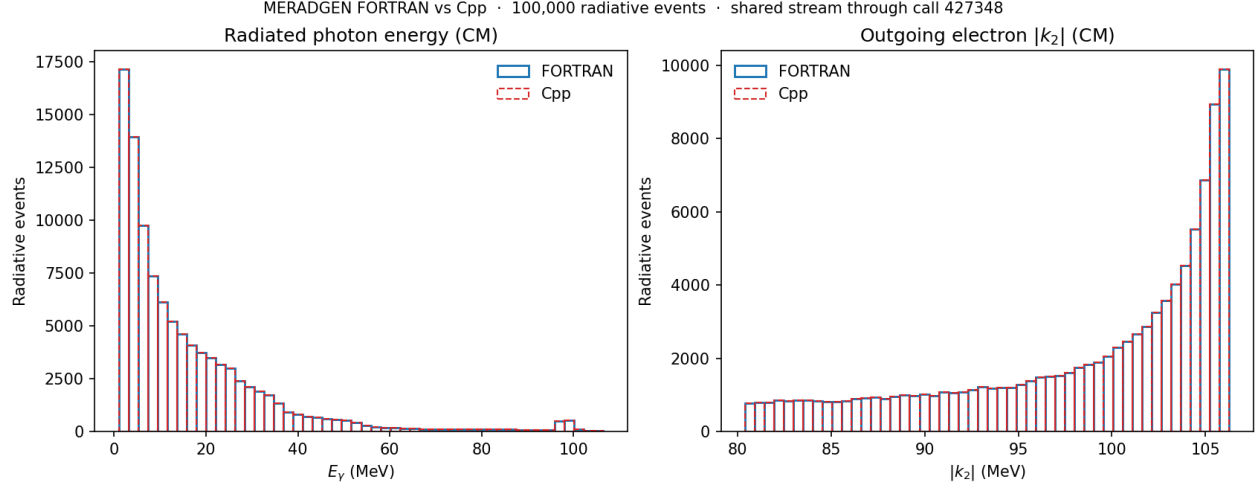


Figure 1. FORTRAN vs. Cpp overlay, 100,000 radiative events, shared stream through call 427,348. Left: E_γ (MeV). Right: $|k_2|$ (MeV).

3 Four-vector deltas (quad plot)

Figure 2 is Cpp minus FORTRAN on the same 100,000 radiative events. Read it left to right, top to bottom:

- (1) **Photon ΔE_γ .** Histogram of the photon energy difference. This panel is *only* energy, not 3-momentum. A difference at 10^{-15} GeV sits in the $\Delta E = 0$ bin at this axis scale. All nonzero ΔE_γ events are listed in Table 1 (up to 20).
- (2) **Photon $\max|\Delta|$ vs. stream event.** For each event, $\max(|\Delta E_\gamma|, |\Delta \vec{p}_\gamma|)$. Points are the events that are not identically zero. Table 2 lists them (up to 20).
- (3) **Electron ΔE_{k_2} .** Histogram of the outgoing-electron energy difference. Same binning caveat as panel (1). All nonzero ΔE_{k_2} events are in Table 3 (up to 20).
- (4) **Electron $\max|\Delta|$ vs. stream event.** $\max(|\Delta E|, |\Delta \vec{k}_2|, |\Delta |k_2||)$. Table 4 lists them (up to 20).

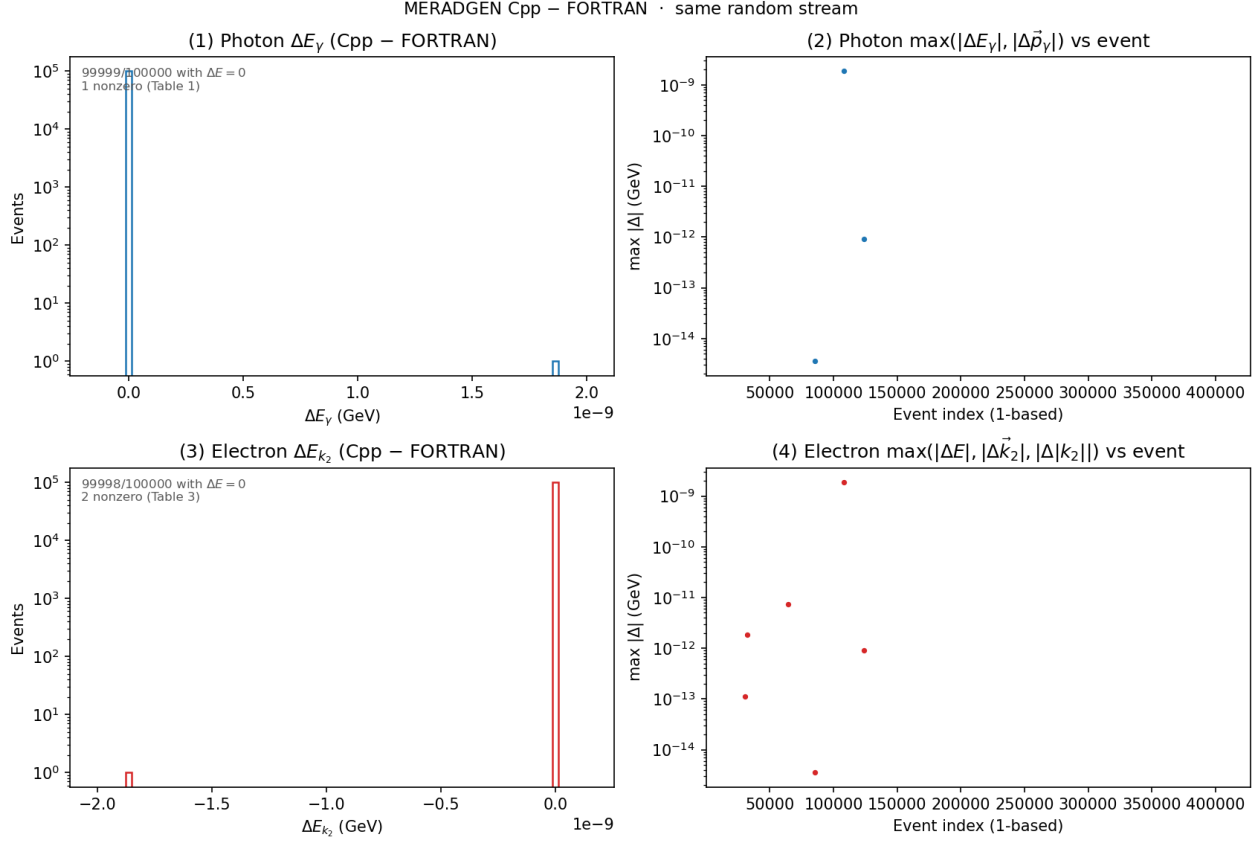


Figure 2. Quad plot of Cpp – FORTRAN, GeV. Horizontal axis on (2) and (4) is the 1-based meradgen call index in `quads.txt`, not the radiative serial number. Annotations count zeros of that panel’s quantity. ΔE is the electron energy difference (the physics quantity); $\Delta |k_2|$ is the change in momentum length. $|\Delta \vec{k}_2|$ is a port-check residual, not a quantity of physics interest. Panel (3) is ΔE only; panel (4) takes the max of all three.

Table 1. Panel (1) — photon events with $\Delta E_\gamma \neq 0$. Events whose photon 3-momentum differs but whose energy matches sit in the $\Delta E = 0$ bin of that histogram; they appear in Table 2. 1 such event in 100,000 (showing 1). Sorted by $|\Delta E_\gamma|$.

Stream event	E_γ FORTRAN (GeV)	E_γ Cpp (GeV)	ΔE_γ (Cpp–FORTRAN)	$ \Delta E_\gamma $
108405	$2.887(10^{-2})$	$2.887(10^{-2})$	$1.863(10^{-9})$	$1.863(10^{-9})$

Table 2. Panel (2) — photon events with any nonzero ΔE_γ or $|\Delta \vec{p}_\gamma|$. 3 such events in 100,000 (showing 3). Sorted by $\max |\Delta|$.

Stream event	E_γ FORTRAN	E_γ Cpp	ΔE_γ	$ \Delta \vec{p}_\gamma $	$\max \Delta $
108405	$2.887(10^{-2})$	$2.887(10^{-2})$	$1.863(10^{-9})$	0	$1.863(10^{-9})$
123695	$1.287(10^{-3})$	$1.287(10^{-3})$	0	$9.095(10^{-13})$	$9.095(10^{-13})$
85445	$2.637(10^{-2})$	$2.637(10^{-2})$	0	$3.553(10^{-15})$	$3.553(10^{-15})$

Table 3. Panel (3) — electron events with $\Delta E_{k_2} \neq 0$. Momentum-only differences sit in the $\Delta E = 0$ bin of that histogram and are in Table 4. 2 such events in 100,000 (showing 2). Sorted by $|\Delta E_{k_2}|$.

Stream event	E_{k_2} FORTRAN (GeV)	E_{k_2} Cpp (GeV)	ΔE_{k_2} (Cpp–FORTRAN)	$ \Delta E_{k_2} $
108405	$9.168(10^{-2})$	$9.168(10^{-2})$	$-1.863(10^{-9})$	$1.863(10^{-9})$
32158	$1.059(10^{-1})$	$1.059(10^{-1})$	$-1.819(10^{-12})$	$1.819(10^{-12})$

Table 4. Panel (4) — electron events with any nonzero $|\Delta E|$, $|\Delta \vec{k}_2|$, or $|\Delta |k_2||$. 6 such events in 100,000 (showing 6). Sorted by $\max |\Delta|$.

Stream event	E_{k_2} FORTRAN	E_{k_2} Cpp	$ \Delta E $	$ \Delta \vec{k}_2 $	$ \Delta k_2 $	$\max \Delta $
108405	$9.168(10^{-2})$	$9.168(10^{-2})$	$1.863(10^{-9})$	0	0	$1.863(10^{-9})$
64502	$9.312(10^{-2})$	$9.312(10^{-2})$	0	$7.276(10^{-12})$	$7.083(10^{-12})$	$7.276(10^{-12})$
32158	$1.059(10^{-1})$	$1.059(10^{-1})$	$1.819(10^{-12})$	0	0	$1.819(10^{-12})$
123695	$1.059(10^{-1})$	$1.059(10^{-1})$	0	$9.095(10^{-13})$	$1.098(10^{-14})$	$9.095(10^{-13})$
30363	$8.286(10^{-2})$	$8.286(10^{-2})$	0	$1.137(10^{-13})$	$1.887(10^{-14})$	$1.137(10^{-13})$
85445	$8.950(10^{-2})$	$8.950(10^{-2})$	0	$3.553(10^{-15})$	$6.106(10^{-16})$	$3.553(10^{-15})$

4 Conclusion

This is the equal comparison: `meradgen-cpp/` is built to match `meradgen-fortran/`, including FORTRAN’s constants and 32-bit four-vectors. On a shared random stream it does so for 100,000 radiative events. Overlay histograms of E_γ and $|k_2|$ are bin-identical. The radiative channel never differs.

Table 5 lists every stream event whose dumped VPRAD or PHIRAD 32-bit four-vector differs. There are six. Each mismatch is exactly one unit in the last place (ULP) of a *single* 32-bit component: the FORTRAN and Cpp values are neighbouring numbers in the IEEE-754² format. Occasional 1-ULP differences of this kind are expected. The two compilers do not always add, multiply, or round in the same internal order, even when the written formulas are the same, so they can land on adjacent representable numbers. None of those six reaches the relative 10^{-6} physics bar. The largest is stream event 108405, $1.863(10^{-9})$ GeV on PHIRAD E (relative $6.5(10^{-8})$), a factor of fifteen below that bar. In hexadecimal³ the bit pattern of that component is `0x3CEC7821` in FORTRAN and `0x3CEC7822` in Cpp, a 1-ULP step.

Table 5. All six stream events with a nonzero dumped 32-bit four-vector. Each row is one neighbouring IEEE-754 encoding (1 ULP). Sorted by $|\Delta|$.

Stream event	Component	FORTTRAN bits	Cpp bits	$ \Delta $ (GeV)	ULPs
108405	PHIRAD E	<code>0x3CEC7821</code>	<code>0x3CEC7822</code>	$1.863(10^{-9})$	1
64502	VPRAD p_x	<code>0xB8F3CA69</code>	<code>0xB8F3CA68</code>	$7.276(10^{-12})$	1
32158	VPRAD E	<code>0xB792D1C4</code>	<code>0xB792D1C3</code>	$1.819(10^{-12})$	1
123695	PHIRAD p_z	<code>0x370C7F1F</code>	<code>0x370C7F1E</code>	$9.095(10^{-13})$	1
30363	VPRAD p_y	<code>0x35CAB579</code>	<code>0x35CAB578</code>	$1.137(10^{-13})$	1
85445	PHIRAD p_y	<code>0x330FA9B2</code>	<code>0x330FA9B3</code>	$3.553(10^{-15})$	1

This does not necessarily certify the production tree `meradgen-cpp-final/`. Another comparison, to ensure that `meradgen-cpp/` and `meradgen-cpp-final/` agree within computational expectations, will need to be performed.

5 Important information

The 100,000-radiative sample is reproduced by

```
python3 comparison/run_compare.py --seed 20260831
--n-radiative 100000 --n-quads 500000 --plot
```

The shared stream itself is written by `comparison/generate_stream.py` (seed 20260831, 500,000 quads) if it is generated separately. Toolchain for this note: `g++/gfortran -O2 -ffp-contract=off -fno-fast-math`.

²IEEE-754 is the usual computer standard for storing real numbers in a fixed number of bits. A 32-bit value (also called float32 or binary32) can only represent certain numbers exactly. The next larger or smaller allowed number is one ULP away: the smallest step that encoding can take at that size.

³A pattern such as `0x3CEC7821` is the 32 bits of the IEEE-754 number written in base 16. Values that differ by one in the last hex digit (here `0x3CEC7821` vs. `0x3CEC7822`) are neighbouring representable floats: one ULP.

A MERADGEN Variables (Code and Math)

Code (meradgen-fortran)	Math	Meaning in this note
ich	—	Channel: 0 = no real photon, 1 = radiative.
vprad	$p_2 - p_1$	CM four-momentum of the scattered electrons, 32-bit. Dumps print this as VPRAD .
phirad	E_γ	CM four-momentum of the real photon, 32-bit. E_γ is its energy. Dumps print this as PHIRAD .
—	k_2	Outgoing beam electron, rebuilt from vprad and phirad .
vpgen	—	Input virtual-photon four-momentum ($k_1 - k_2$) in the CM.
vgen, t1gen, zgen	—	Internal double-precision kinematics. Not the official pass/fail.
pl	P_L	Longitudinal beam polarisation (here -1). Argument of meradgen .
elab	E_{lab}	Lab beam energy (here 45 GeV).
thetacm	θ_{CM}	CM polar angle (here 90°).
phi	ϕ	CM azimuthal angle (here 10°).
urand	quad	FORTTRAN RNG. The shared-stream harness replaces each urand(iy) with r1-r4 .